

BUYERS' GUIDE

Selecting The Right POWER SUPPLIES For ELECTRONIC DEVICES

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A power supply converts AC voltage into regulated DC voltage. Read on to find out how to select the best power supply for an electronic device

A power supply for an electronic equipment is a circuit that converts AC voltage into regulated DC voltage. An electronic device does not normally use electric energy in the form in which it is produced or distributed. It requires some form of energy conversion. A power supply unit (PSU) transfers electric energy from a given source to a given load using electronic circuits. There are different types of PSUs depending on their mode of operation and application.

Basic building components

A PSU derives power from the AC mains supply and performs many tasks before transferring it to the output electronic device. These tasks include:

- Changing the level of supply to a value suitable for driving the load circuit.
- Producing DC supply from the mains supply AC sine wave.
- Preventing any AC signal at supply output.
- Ensuring constant output voltage independent of changes in AC supply voltage, load current and temperature.

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To do these things, a basic PSU has four main constituents: transformer, rectifier, filter and regulator circuit.

Input transformer. This is used to transform incoming line voltage (AC) down to the required (AC) level for the power supply. It isolates the output circuit from the line supply.

Rectifier. This converts incoming AC signal into DC signal.

Filter. This is necessary to ensure

